Homework 1 - 09/08 Content

MATH 2263, Garrett Credi

09/08/2026

Problems listed as AC are found in the Active Calculus textbook at the end of the section listed.

Problems listed as APEX are found in the APEX Calculus Textbook again at the end of the listed section.

Problems listed as PAULS-PRACT are found in Paul's Online Math Notes - Review Problems while problems listed as PAULS-ASSIGN are found in Paul's Online Math Notes - Assignment Problems and are indexed via [Topic]-[Section]-[Problem Number]

1 Type 1 - Functions

Choose three of the following problems to solve:

1. PAULS-ASSIGN Review.Functions.1-13 (each of these counts as one problem)

2 Type 2 - Solving Equations

Choose three of the following problems to solve:

1. PAULS-ASSIGN.Review.Functions.14-21 (each of these counts as one problem)

3 Type 3 - Function Composition

Choose two of the following problems to solve:

1. PAULS-ASSIGN.Review.Functions.52-55, also identify the domain and range of the composites you find (each of these count as one problem)

4 Type 4 - Function Graphs

Choose three of the following problems to solve:

1. PAULS-ASSIGN.Review.Common Graphs.1-26 (each of these counts as one problem)

5 Type 5 - Trig Functions

Choose three of the following problems to solve:

1. PAULS-ASSIGN.Review.Trig Functions.1-19 (each of these counts as one problem)

6 Challenge Problem

Suppose that you are running experiment where a population of bacteria grows according to the following function

$$P(t) = e^t + 10$$

where t is the time since population introduction and $P(t)$ is the organic mass of the population measured in grams.

Additionally, suppose that the bacteria produce/consume energy at a level depending on their total organic mass according to

$$E(p) = 10\sin^2(p)$$

where p is the organic mass of the population in grams, and E is energy in Joules.

If your instruments measure that the bacteria are producing exactly 5 Joules of energy, what is the earliest possible time since population introduction that the experiment could have been running?